Quantum Resonance Unified Model (QRUM)

Consolidated Summary

Version 2 — Scientific Revision

P. Gabor

Independent Researcher

Contact: p.gabor8107@gmail.com

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Accompanying code: https://github.com/gabor8107/QRUM

Abstract

The Quantum Resonance Unified Model (QRUM) is presented as a phenomenological research programme that organises experimentally testable investigations of residual structure, statistical correlation and environmental dependence in high-precision observational data. Rather than replacing General Relativity, Quantum Mechanics or the Standard Model, QRUM provides a structured methodological framework for reproducibility, transparent analysis and explicit falsifiability.

This consolidated summary reviews the current components of the programme, their interrelations, scientific boundaries and future experimental directions. Individual technical results remain in their respective publications; the present document supplies the overarching scientific orientation.

Keywords: phenomenological modelling; residual analysis; gravitational waves; precision measurement; reproducibility; open science

1. Introduction

Recent advances in gravitational-wave astronomy, optical atomic clocks, atom interferometry and precision metrology have created opportunities for testing subtle observational effects. QRUM was developed as a phenomenological framework intended to organise such investigations into a coherent research programme.

Instead of proposing a replacement for existing theories, the framework emphasises reproducibility, transparency and experimentally testable hypotheses. Each component of the programme can be refined, replaced or rejected without affecting the overall research methodology.

2. Scientific Philosophy

QRUM adopts a conservative scientific position. General Relativity, Quantum Mechanics and the Standard Model remain the reference descriptions of established physics. Phenomenological response functions are introduced only as observational modelling tools whose usefulness must ultimately be determined through experiment.

Independent verification, statistical robustness and methodological transparency are considered essential components of every publication in the programme.

3. Why a Phenomenological Framework?

Phenomenological models provide practical descriptions of observable behaviour without requiring a complete microscopic theory. Within QRUM this strategy allows individual hypotheses to be tested independently while avoiding unnecessary theoretical assumptions.

No generally covariant gravitational action is proposed. Frequency-dependent response functions are treated exclusively as spectral or statistical test functions, not as fundamental modifications of the Einstein field equations.

4. Structure of the QRUM Research Programme

The current programme consists of independent but complementary publications:

- QRUM Phase 1 – Residual Framework
- Correlation-Based Framework
- Environmental Dependence of Compact Object Frequency Populations
- Astrophysical Frequency Survey
- Black Hole Resonance Dynamics (exploratory)
- Multi-Modal Detection Architecture (roadmap)

Together these publications define the present scope of the QRUM research programme and provide the foundation for future experimental investigations.

5. Scope of this Review

This document serves as the central overview of the QRUM programme. Individual technical details remain within their respective publications, while the present review focuses on the

scientific motivation, conceptual relationships between studies, methodological consistency and long-term experimental roadmap.

6. QRUM Phase 1 – Residual Framework

The first major component of the QRUM research programme investigates whether small residual structures remain after standard gravitational-wave signal modelling. The purpose is methodological rather than declarative: to establish a reproducible workflow for analysing residuals and assessing whether additional phenomenological descriptions warrant further investigation.

Scientific Motivation

Residual analysis has become an important diagnostic tool because it allows researchers to examine the agreement between observational data and reference models. Within QRUM, residual studies are treated as exploratory analyses designed to identify patterns that can later be tested using independent datasets and improved statistical methods.

Methodological Overview

The framework emphasises transparent preprocessing, consistent data selection, reproducible computational workflows and explicit reporting of assumptions. Any candidate residual feature should be evaluated against instrumental artefacts, detector noise, environmental influences and statistical fluctuations before interpretation.

Interpretation Strategy

QRUM deliberately avoids interpreting phenomenological fits as evidence for new physical interactions. Instead, fitted response functions are regarded as descriptive tools that may help organise future investigations. Independent replication is considered essential before any broader conclusions can be drawn.

Current Status

Application to public LIGO strain data has yielded null or inconclusive results under the present analysis choices. No statistically robust, detector-coherent residual resonance is claimed.

7. Correlation-Based Framework

The Correlation-Based Framework extends the methodological principles introduced in the Residual Framework by examining whether statistically meaningful relationships can be identified across independent observations. Its purpose is not to demonstrate new physics but to establish a reproducible statistical methodology for comparing multiple datasets.

Scientific Motivation

Independent observations provide stronger evidence than isolated measurements. The framework therefore investigates cross-comparison strategies that reduce dependence on any single detector, observation period or processing pipeline. Agreement across independent datasets is treated as a stronger indicator than isolated features.

Methodological Objectives

The primary objective is to develop reproducible statistical procedures for evaluating consistency between observations. Candidate analyses should document preprocessing, parameter choices, uncertainty estimates and alternative explanations. Statistical significance should always be interpreted conservatively.

Role within the Programme

Together with the Residual Framework, the Correlation-Based Framework establishes the statistical foundation upon which later astrophysical, environmental and multi-modal investigations are organised.

8. Environmental Dependence

The Environmental Dependence study examines whether external conditions and Galactic environment can influence the statistical properties of compact-object frequency populations. Public catalogues (primarily ATNF) are combined with standard dark-matter halo models (NFW, Einasto) to test for population-level associations.

Methodological Importance

Environmental characterisation forms a complementary layer of the QRUM programme. Detector configuration, calibration status, ambient conditions and known instrumental behaviour should be evaluated together with the observational data. Explicit documentation of these factors improves transparency and facilitates independent replication.

Current Status

A statistically detectable but very weak population-level association between pulsar spin frequency and model-estimated Galactic dark-matter density has been reported (Spearman $\rho \approx 0.054$). The correlation remains stable under bootstrap and permutation testing yet weakens substantially within subclasses. These results do not provide evidence that dark matter directly organises pulsar frequencies, nor do they validate QRUM as a physical theory.

9. Astrophysical Frequency Survey

The Astrophysical Frequency Survey expands the programme beyond individual observations by examining characteristic frequency ranges reported across different astrophysical systems. The objective is to organise observational information systematically and identify recurring frequency domains that may warrant further investigation.

The survey is descriptive rather than explanatory. It does not assume a common physical origin for similar frequencies but provides a structured observational catalogue that may guide future statistical analyses.

10. Black Hole Resonance Dynamics

The Black Hole Resonance Dynamics study explores phenomenological descriptions of compact-object merger observations. Rather than proposing modifications to General Relativity, this work investigates whether resonance-inspired observational parameterisations can provide an additional descriptive framework for analysing post-merger behaviour and characteristic frequency evolution.

The emphasis remains methodological. Compact-object observations represent valuable datasets for testing statistical consistency, observational reproducibility and phenomenological modelling under extreme physical conditions.

11. Multi-Modal Detection Architecture

The Multi-Modal Detection Architecture represents the experimental roadmap of the QRUM programme. Its objective is to outline how independent precision instruments may collectively investigate phenomenological hypotheses using complementary observational techniques.

Instead of relying on a single experimental platform, QRUM proposes that future investigations should compare results obtained from multiple independent technologies. Candidate platforms include gravitational-wave detectors, optical atomic clocks, atom interferometers, superconducting resonators, precision timing systems and other high-coherence instruments.

Agreement between independent technologies would strengthen confidence in observational conclusions, while disagreement would motivate further methodological investigation.

12. Experimental Roadmap

The long-term objective of the QRUM programme is to encourage systematic experimental testing of its phenomenological hypotheses. The framework is intentionally organised as a sequence of increasingly demanding investigations, beginning with reproducible analyses of existing public datasets and progressing toward dedicated precision experiments.

Near-Term Priorities

The first stage focuses on analyses that can be reproduced using openly available observational data. Independent processing pipelines, cross-validation and sensitivity studies should be encouraged before broader physical interpretation is considered.

Medium-Term Objectives

As the programme matures, coordinated analyses using multiple independent observational platforms become increasingly important. Agreement across different experimental techniques would provide stronger support for methodological consistency than any single dataset alone.

Long-Term Platforms

Future investigations may involve gravitational-wave observatories, optical atomic clocks, atom interferometers, superconducting resonators, SQUID magnetometers, precision timing systems and other high-coherence sensing technologies. These remain prospective; no claim is made that current data require them.

13. Scientific Position and Boundaries

QRUM is presented as a phenomenological research programme. It is intended to organise experimentally testable hypotheses and reproducible analyses rather than replace established theories of physics. Throughout the programme, General Relativity, Quantum Mechanics and the Standard Model remain the reference descriptions against which phenomenological investigations are evaluated.

Explicit boundaries:

- No generally covariant gravitational action is proposed.
- No modification of the Einstein field equations is asserted.
- Frequency-dependent factors are spectral or statistical test functions only.
- Null and inconclusive results are reported as such.

14. Methodological Limitations

Every phenomenological framework has limitations. QRUM does not claim that descriptive agreement alone demonstrates new physical mechanisms. Statistical uncertainty, detector characteristics, environmental influences, model assumptions and computational choices must all be examined before broader conclusions are considered. Independent replication is regarded as essential.

15. Falsifiability

A central objective of QRUM is to formulate hypotheses that are explicitly testable. If future observations consistently fail to reproduce predicted phenomenological behaviour within experimental sensitivity, the corresponding models should be revised or rejected. Negative results therefore contribute directly to scientific progress by constraining the range of viable hypotheses.

16. Reproducibility and Open Science

Open documentation, transparent methodology, public datasets and independent software implementations are encouraged throughout the QRUM programme. Analysis code is released under the MIT License; manuscripts are deposited with DOI on Zenodo. These principles allow external researchers to evaluate, reproduce and critically assess each component independently.

17. Position within Precision Physics

QRUM should be interpreted as a precision-physics research strategy that complements existing experimental programmes. Its long-term value depends on careful methodology, conservative interpretation and continued comparison with independent observations.

18. Conclusions

The Quantum Resonance Unified Model (QRUM) has been presented throughout this review as a structured phenomenological research programme rather than a replacement for established physical theories. Its primary contribution is the organisation of multiple independent investigations into a coherent framework centred on reproducibility, precision measurements and explicit experimental testing.

The publication series demonstrates how residual analysis, statistical correlation, environmental characterisation, astrophysical surveys and multi-modal experimental concepts can be developed as complementary components of a unified research strategy. Each publication remains scientifically independent while contributing to a common methodological philosophy.

19. Future Research Priorities

Future work should emphasise independent replication, improved statistical methods, expanded public datasets and additional precision-measurement platforms. The programme is expected to evolve incrementally, with individual phenomenological components being refined, confirmed or rejected according to experimental evidence.

Long-term objectives include broader comparisons across observational domains, improved uncertainty quantification and closer integration of independent experimental results.

20. Publication Strategy

Maintaining separate technical publications alongside a consolidated review provides a clear scientific structure. Individual manuscripts preserve methodological detail, whereas the present summary offers a unified overview of the overall research programme and its current status.

Final Statement

The QRUM programme should ultimately be evaluated by the quality of its methodology and by the outcome of future experimental tests. Regardless of whether individual hypotheses are supported or excluded, the framework is intended to encourage transparent, reproducible and falsifiable investigations within precision physics.

Data and Code Availability

Analysis code: https://github.com/gabor8107/QRUM (MIT License).

Manuscripts and versions are archived on Zenodo.

Primary public data sources include the Gravitational Wave Open Science Center (GWOSC) and the ATNF Pulsar Catalogue.

References

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Appendix A — Overview of the Current QRUM Publication Series

Table 1 summarises the principal components of the QRUM programme, their primary objectives, data scope, methodology and current scientific status.

Publication	Primary Objective	Main Data / Scope	Methodology	Current Status
Residual Framework	Residual analysis	Public GW observations	Reproducible residual workflow	Null / inconclusive
Correlation-Based Framework	Cross-dataset consistency	Independent observations	Statistical correlation	Methodological framework
Environmental Dependence	Environmental influences	Astrophysical catalogues	Population statistics	Exploratory (weak signal)
Astrophysical Frequency Survey	Frequency catalogue	Published astrophysical frequencies	Comparative survey	Descriptive
Black Hole Resonance Dynamics	Compact-object interpretation	Merger observations	Phenomenological analysis	Exploratory
Multi-Modal Detection Architecture	Future experiments	Multiple precision platforms	Integrated roadmap	Conceptual

Appendix B — Publication Timeline

The QRUM programme has been developed through a sequence of independent publications, each addressing a specific methodological question.

- Phase 1: Establishment of the Residual Framework.
- Phase 2: Development of statistical correlation methodology.
- Phase 3: Investigation of environmental dependence.
- Phase 4: Compilation of the Astrophysical Frequency Survey.
- Phase 5: Exploratory compact-object resonance study.
- Phase 6: Proposal of the Multi-Modal Detection Architecture.
- Phase 7: Consolidation of the publication series within the present Summary (Version 2).

Future revisions may extend this timeline as additional software releases, independent replications and experimental investigations become available.

Appendix C — Glossary

Phenomenological Model: A descriptive framework intended to organise observable behaviour without claiming a fundamental physical theory.

Residual Analysis: Examination of the difference between observational data and a reference model.

Correlation: Statistical association between independent datasets or measurements.

Reproducibility: The ability of independent researchers to repeat an analysis using the same methods and data.

Falsifiability: The property that a hypothesis can be tested and potentially rejected by observation.

Precision Physics: Research based on extremely accurate measurements used to test physical models.

Multi-Modal Detection: Use of multiple independent experimental platforms to investigate the same hypothesis.

Null / Inconclusive Result: An outcome in which the data do not support a detection claim under the stated analysis choices; scientifically informative for constraining hypotheses.

Appendix D – Structure of the QRUM Research Programme

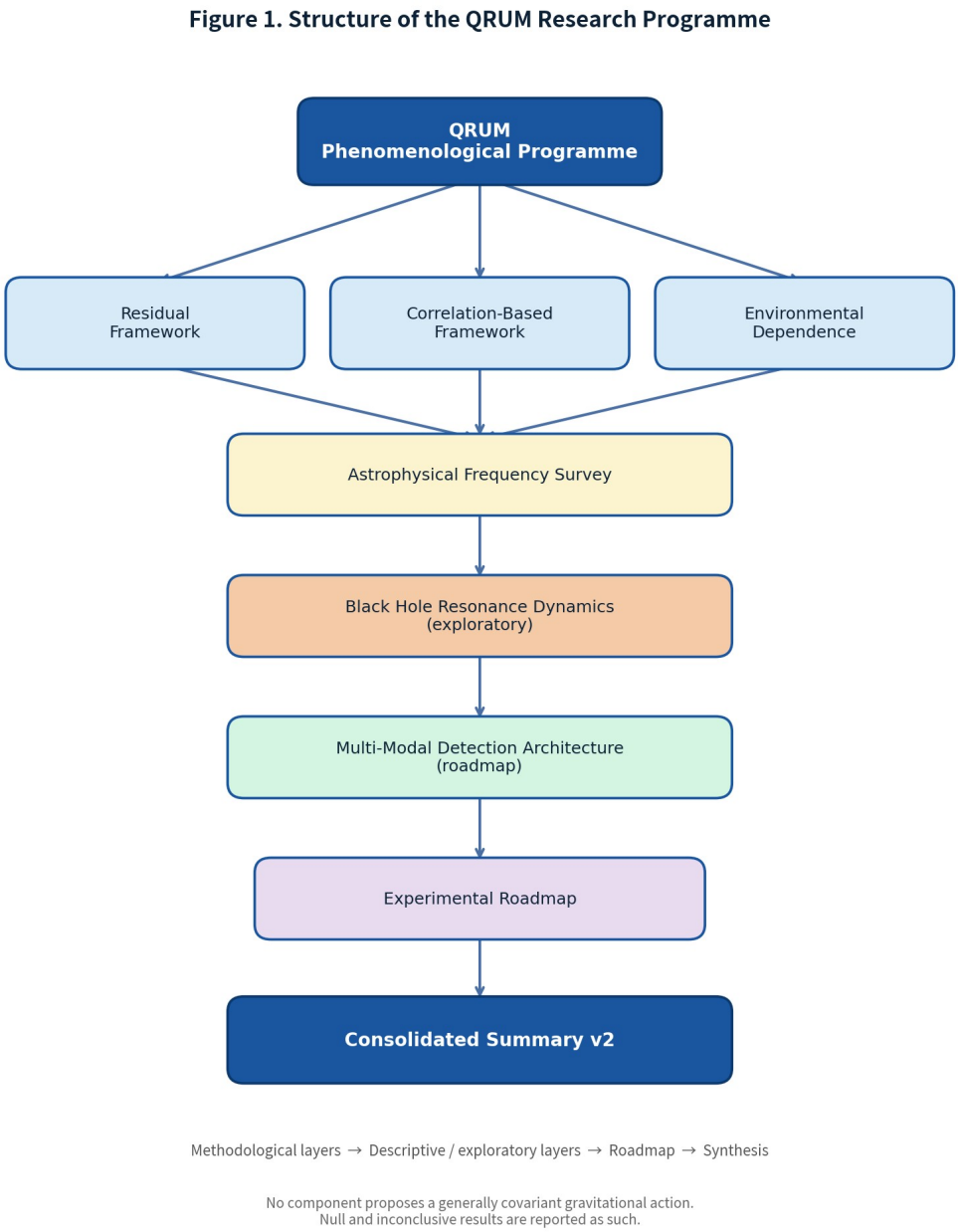


Figure 1. Structure of the QRUM Research Programme.